

Motus: A Unified Latent Action World Model

Hongzhe Bi^{1*†}, Hengkai Tan^{1*†}, Shenghao Xie^{2,1*}, Zeyuan Wang^{1*}, Shuhe Huang^{1*}, Haitian Liu^{1*},
Ruowen Zhao¹, Yao Feng¹, Chendong Xiang¹, Yinze Rong¹, Hongyan Zhao¹, Hanyu Liu²,
Zhizhong Su³, Lei Ma², Hang Su¹, Jun Zhu¹

¹Dept. of Comp. Sci. and Tech., Institute for AI, BNRist Center, THBI Lab,
Tsinghua-Bosch Joint ML Center, Tsinghua University

²Peking University ³Horizon Robotics

*Joint first authors [†]Joint project lead

{bhz24, thj23}@mails.tsinghua.edu.cn, dcszj@tsinghua.edu.cn

Project Page: <https://motus-robotics.github.io/motus>

Abstract

While a general embodied agent must function as a unified system, current methods are built on isolated models for understanding, world modeling, and control. This fragmentation prevents unifying multimodal generative capabilities and hinders learning from large-scale, heterogeneous data. In this paper, we propose **Motus**, a unified latent action world model that leverages existing general pretrained models and rich, sharable motion information. Motus introduces a Mixture-of-Transformer (MoT) architecture to integrate three experts (i.e., understanding, video generation, and action) and adopts a UniDiffuser-style scheduler to enable flexible switching between different modeling modes (i.e., world models, vision-language-action models, inverse dynamics models, video generation models, and video-action joint prediction models). Motus further leverages the optical flow to learn latent actions and adopts a recipe with three-phase training pipeline and six-layer data pyramid, thereby extracting pixel-level “delta action” and enabling large-scale action pretraining. Experiments show that Motus achieves superior performance against state-of-the-art methods in both simulation (a **+15%** improvement over X-VLA and a **+45%** improvement over $\pi_{0.5}$) and real-world scenarios (improved by **+11~48%**), demonstrating unified modeling of all functionalities and priors significantly benefits downstream robotic tasks.

1. Introduction

A unified model is essential for embodied agents to integrate a spectrum of cognitive functions—from understanding scenes and instructions, imagining possible futures, to

predicting consequences and generating actions—into a unified whole. However, existing methods model these capabilities in isolation: some rely on vision-language-action models (VLAs) [5, 8, 11, 26, 31, 36, 60, 65] to learn static policies from vision and language; others use world models or generative approaches built on predicted futures [4, 7, 19, 21, 25, 28, 39, 41, 53, 56, 62]; and $\mathcal{F}_1$ [32] combines VLAs and inverse dynamics models (IDMs) by explicitly imagining future visual observations, but it excludes world models or video generation models (VGMs), resulting in incomplete unification. These approaches fragment what should be a unified system into 5 separate modeling tasks:

- VLA: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- WM: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{a}_{t+1:t+k})$.
- IDM: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_{t:t+k})$.
- VGM: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- Video-Action Joint Prediction Model:
 $p(\mathbf{o}_{t+1:t+k}, \mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.

Two fundamental challenges (detailed in Sec. 3) hinder the integration of these capabilities. First, *unifying such multimodal generative capabilities* within one framework is nontrivial. While unified world models (UWMs) [64] offer a theoretical prototype, they are typically trained from scratch or with limited priors, lacking either robust vision-language understanding from vision-language models (VLMs) or rich physical interaction knowledge from VGMs. Second, embodied intelligence demands the ability to *learn from large-scale heterogeneous data*—including internet videos, ego-centric human demonstrations, and multi-robot trajectories—but action spaces vary widely across embodiments, and most video data lack action labels, making it difficult to pretrain action experts with general motion and interaction priors.

To address these challenges, we propose **Motus**, a unified

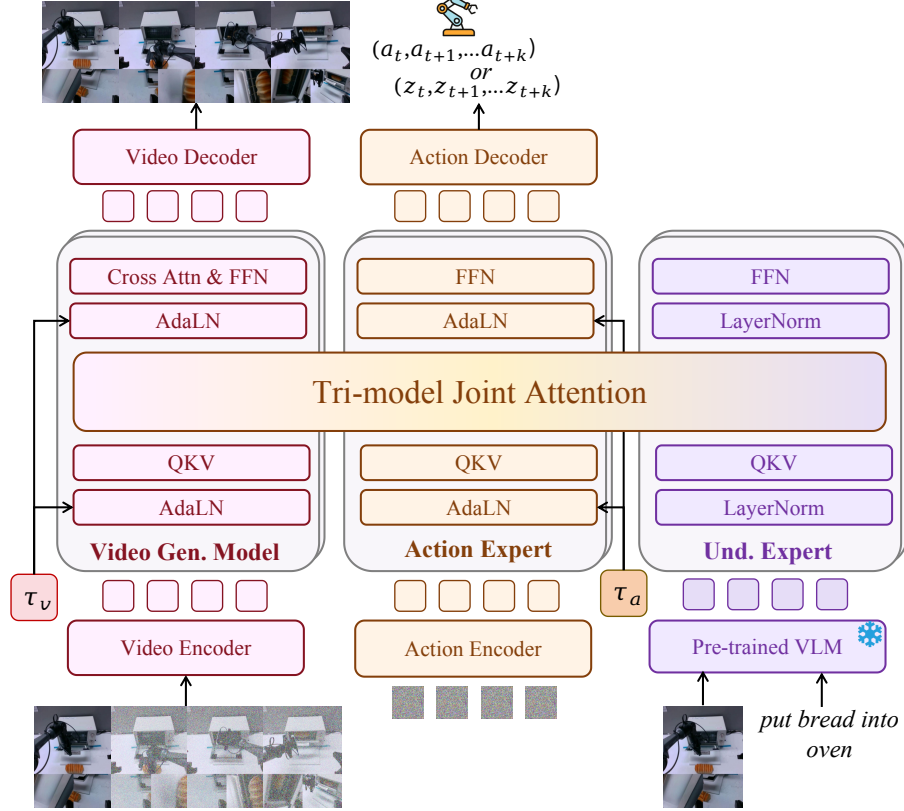


Figure 1. Motus Architecture. Here, $a_t \dots a_{t+k}$ are actions, $z_t \dots z_{t+k}$ are latent actions, and τ_v and τ_a are the rectified flow timesteps for the video generation model and the action expert, respectively.

latent action world model that integrates pretrained experts within a Mixture-of-Transformers (MoT) architecture. Our approach unifies the 5 key distributions by connecting a video generator (generative expert), an action expert, and a vision-language understanding expert via shared multi-head self-attention layers—a design we term **Tri-model Joint Attention**—which preserves specialized functionalities while enabling cross-modal knowledge fusion. To further coordinate multimodal generation, Motus incorporates a UniDiffuser-like scheduler, allocating distinct timesteps and noise scales to each modality (*e.g.*, videos and actions). This enables a unified manner for simultaneous modeling marginal, conditional, and joint distributions, as well as adaptive switching among different inference modes (*e.g.*, VLA, WM, IDM, VGM, Video-Action Joint Prediction Model).

Additionally, to leverage heterogeneous data at scale, we introduce latent actions, which encode motion patterns from optical flow as a pixel-level “delta action”. This representation bridges visual dynamics with control signals, enabling the action expert to be pretrained on diverse unlabeled videos and robot trajectories. Specifically, a pretrained deep compression autoencoder (DC-AE) with additional lightweight downsampling modules is used to reconstruct optical flow,

whereas its encoded low-dimensional latents are supervised with a few action labels, both task-related and task-agnostic, thus steering the focus towards patterns associated with robotic activities.

Subsequently, Motus undergoes a three-phase pretraining–finetuning pipeline (*i.e.*, video pretraining, latent action pretraining, and embodiment-specific action finetuning) on a six-layer data pyramid spanning web-scale, egocentric human, simulation, task-agnostic, multi-robotic, and target-robotic data. This recipe aligns behaviors across different embodiments within the motion space described by optical flows and shares such interaction knowledge with target embodiments to enhance the generalization in downstream tasks, thereby providing the action expert with pretraining like other experts.

Overall, our contributions can be summarized as follows:

- A unified embodied foundation model that integrates five mainstream paradigms (*i.e.*, WMs, IDMs, VLAs, VGMs, and Video-Action Joint Prediction Models) without compromising general multimodal priors.
- A scalable robotic recipe with a three-phase training pipeline and six-layer data pyramid that leverages optical flow-based latent action to learn cross-embodiment

transferable motion knowledge.

- Extensive experiments show that Motus significantly outperforms state-of-the-art approaches in both simulation (a **+15%** improvement over X-VLA [60] and a **+45%** improvement over $\pi_{0.5}$ [8]) and real-world scenarios (improved by **+11~48%**), demonstrating that large-scale general and domain-specific priors can be effectively fused to enhance the generalization of policy learning.

2. Related Works

2.1. Unified Multimodal Models

Unified multimodal models jointly model various modalities and tasks within a single generative framework [29, 40, 45, 47, 49, 52], showing broad applications across several domains [35, 54, 63]. In particular, Bagel [18] achieves unification via MoT [30], sharing the multi-head self-attention layers between understanding experts and generation experts. In contrast, existing embodied foundation models are developed independently, spawning multiple disparate paradigms: some leverage the text-image understanding capabilities of VLMs to learn action prediction [6, 8, 27], while others utilize VGMs to generate video sequences and infer actions from consecutive frames [19, 21, 62]. Recently, $\mathcal{F}_1$ [32] extends VLAs to explicitly imagine future visual states and output actions by IDMs, thereby merging both models. Furthermore, UWM [64] unifies WMs, VLAs, IDMs, VGMs, and Video-Action Joint Prediction Models within a single diffusion backbone, making an initial exploration of complete robotic models. Unlike UWM, our method goes beyond unified modeling by further incorporating internet-scale general multimodal priors and specialized priors from massive robotic trajectories.

2.2. Latent Action Models

Latent actions mitigate the scarcity of action labels by capturing visual dynamics, and are typically derived by coupling IDMs with forward dynamics models (FDMs) to reconstruct the next frame conditioned on the previous one [9, 10, 20, 37]. Initially, RGB images are used for supervision, but this introduces task-irrelevant appearance information [58]. To remove such interference, a common approach is restricting autoencoder’s capacity to encode low-dimensional latents [15, 38, 55], thereby reducing the inclusion of redundancy. AdaWorld [22] attempts to decouple the representations, such as β -VAE [23], in order to retain only the useful factors. Other approaches explore alternative reconstruction objectives, *e.g.*, DINOv2 features [11, 15, 50], object keypoints [17, 51, 57], and language instructions [16], which carries rich semantic and spatial features. Moreover, LAOM [34] employs a few action labels to encourage the model to focus on robotic activities. Building on these advances and inspired by optical flow as a universal motion

expression [12, 46, 61], we use it to align cross-embodiment behaviors and learn latent actions to facilitate large-scale pretraining.

3. Problem Formulation and Challenges

Embodied Policies We consider the task of language-conditioned robotic manipulation. For each embodiment, the task defines an action $\mathbf{a} \in \mathcal{A}$, an observation $\mathbf{o} \in \mathcal{O}$ (visual input), a language instruction $\ell \in \mathcal{L}$, and the proprioception of the robot $\mathbf{p}$, where $\mathcal{A}$, $\mathcal{O}$ and $\mathcal{L}$ denote the action space, the observation space, and the language instruction space respectively. The task typically provides an expert dataset $D_{\text{expert}} = \{\{\ell, \mathbf{p}_1, \mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{p}_N, \mathbf{o}_N, \mathbf{a}_N\}\}$, which contains robot proprioception, visual observations, and actions collected by an expert over N timesteps, along with corresponding language annotations for each trajectory. We train a policy parameterized by θ on D_{expert} . At each timestep t , the policy predicts the next k actions (action chunking [59]) based on the current observation and proprioception, modeling the distribution $p_{\theta}(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{p}_t, \ell)$ or $p_{\theta}(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$. The policy p_{θ} is trained to maximize the likelihood objective:

$$\max_{\theta} \mathbb{E}_{(\mathbf{o}_t, \mathbf{p}_t, \mathbf{a}_{t+1:t+k}, \ell) \sim D_{\text{expert}}} \log p_{\theta}(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{p}_t, \ell). \quad (1)$$

Furthermore, based on the symbolic definitions above, we can derive the probability distributions for the 5 modeling types of embodied intelligence, which can be integrated into a single model for training:

- VLA: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- WM: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \mathbf{a}_{t+1:t+k})$.
- IDM: $p(\mathbf{a}_{t+1:t+k} \mid \mathbf{o}_{t:t+k})$.
- VGM: $p(\mathbf{o}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.
- Video-Action Joint Prediction Model:
 $p(\mathbf{o}_{t+1:t+k}, \mathbf{a}_{t+1:t+k} \mid \mathbf{o}_t, \ell)$.

Challenge 1: Unifying Multimodal Generative Capabilities. A capable embodied agent must integrate a spectrum of cognitive functions—from understanding scenes and instructions, imagining possible futures, to predicting consequences and generating actions—to possess a human-like capacity, as a unified whole. Current models, however, are fragmented and fail to capture the full set of necessary capabilities within one system. This presents a challenge: how to unify the modeling of five key distributions—VLA, World Model, IDM, Video Generation Model, and Video-Action Joint Prediction Model—within a single framework. While prior work, such as UWMs [64], has made some progress, a critical limitation persists: these approaches are either trained from scratch, built upon smaller base models, or—even when incorporating some priors—invariably lack the full spectrum of knowledge, missing *either* visual understanding priors from VLMs *or* physical interaction priors

from VGMs. Consequently, they lack the comprehensive world knowledge required for robust and generalizable embodied intelligence. Therefore, the nontrivial challenge of jointly modeling various distributions of vision, language, and action within a unified framework remains unaddressed, which is precisely the gap our work fills.

Challenge 2: Utilization of Heterogeneous Data. A central challenge in embodied intelligence is how to make effective use of large scale heterogeneous data. Action spaces vary widely between embodiments in dimension, range, and semantics, and robots differ in morphology, actuation, and sensing. As a result, control signals are not directly reusable and policies struggle to learn universal priors that transfer across embodiments. Existing approaches, including [8, 31, 43, 60], try to address this by using a general backbone with embodiment-specific information injection, or constructing high-dimensional action vectors that forcibly unify different embodiments. However, they still depend primarily on labeled robotic trajectories and cannot integrate these datasets with large-scale internet videos or egocentric human videos, which lack action annotations but contain abundant motion and physical interaction cues. This limitation prevents large-scale pretraining of the action expert and reduces the ability to learn general motion priors.

4. Methodology

4.1. Motus

Model Architecture. To address the challenges of unifying multimodal generative capabilities outlined in Sec. 3, we propose Motus, a unified latent action world model. First, Motus is designed as a general generative model that jointly learns on heterogeneous multimodal data, thereby integrating the diverse capabilities (*e.g.*, modeling 5 distributions) of a general-purpose system within a single network. Second, to circumvent the need for impractical amounts of aligned multimodal data, Motus leverages the rich, pre-trained priors of existing foundation models. It integrates a pretrained VGM (generative expert), an understanding expert with pretrained VLM, and an action expert within a Mixture-of-Transformers (MoT) architecture (as shown in Fig. 1), effectively fusing their complementary strengths—encompassing scenes understanding, instructions interpreting, consequences prediction, future video imagination, and action planning—without requiring full end-to-end training from scratch. Unlike Unified World Models (UWMs) [64], which simply concatenate observation tokens and action tokens and process them through a single series of N UWM blocks (containing self-attention and feed-forward network (FFN) layers), our approach leverages pretrained VLMs and VGMs by adopting a MoT structure. In our model, each expert maintains an individual Transformer module, while

the multi-head self-attention layers are concatenated, *i.e.*, **Tri-model Joint Attention**. This not only preserves distinct function roles across experts without causing task interference but also enables effective cross-modal feature fusion, encouraging diverse pretrained knowledge to complement one another. During training, Motus jointly predicts chunks of videos and actions with rectified flow-based objectives:

$$l_{\text{action}}^{\theta} = \mathbb{E}_{(\mathbf{o}_{t:t+k}, \mathbf{a}_{t+1:t+k}, \ell) \sim \mathcal{D}} \left\| v_a^{\theta} - (\epsilon_a - \mathbf{a}_{t+1:t+k}) \right\|_2^2, \\ \tau_a \sim \mathcal{U}(0, T_{\tau}) \\ \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$l_{\text{obs}}^{\theta} = \mathbb{E}_{(\mathbf{o}_{t:t+k}, \mathbf{a}_{t+1:t+k}, \ell) \sim \mathcal{D}} \left\| v_o^{\theta} - (\epsilon_o - \mathbf{o}_{t+1:t+k}) \right\|_2^2, \\ \tau_o \sim \mathcal{U}(0, T_{\tau}) \\ \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$l^{\theta} = l_{\text{action}}^{\theta} + l_{\text{obs}}^{\theta}.$$

where $\mathbf{o}_t$ is the condition frame, $\mathbf{o}_{t+1:t+k}$, $\mathbf{a}_{t+1:t+k}$ are subsequent observations and actions, τ_a and τ_o are the assigned timesteps, ϵ_a , ϵ_o are the sampled Gaussian noises, v_a^{θ} , v_o^{θ} are velocity field predicted by our unified model, and $l_{\text{action}}^{\theta}$, l_{obs}^{θ} are loss of observations and actions. By allocating different timesteps and noise scales to videos and actions, respectively, Motus establishes a UniDiffuser-like scheduler to capture heterogeneous data distributions and adaptively switch between various embodied foundation models during inference (*e.g.*, VLA, World Model, IDM, VGM, Joint Prediction). The resulting model understands scenes, follows instructions, predicts outcomes, imagines futures, and outputs actions—all within a unified multimodal architecture.

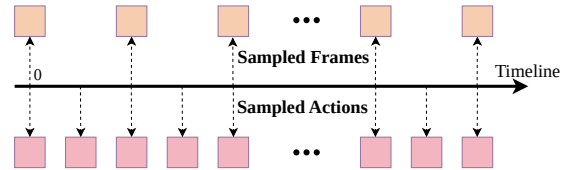


Figure 2. Action-Dense Video-Sparse Prediction. The sampling rates for video frames and actions differ.

Action-Dense Video-Sparse Prediction. Since our model builds upon the widely cited action-chunking technique, Motus needs to predict a chunk of future video and action sequences $\mathbf{o}_{t+1:t+k}$, $\mathbf{a}_{t+1:t+k}$. This leads to several issues: (1) low training and inference efficiency, (2) redundant video frame predictions, and (3) an imbalance in the Tri-modal Joint Attention mechanism—where the number of video tokens significantly exceeds that of action tokens. This imbalance causes the model to overfit to video prediction, thereby weakening its action prediction capability. To address these problems, we propose an Action-Dense Video-Sparse Prediction strategy, as shown in Fig. 2. During both training and

inference, we downsample the video frames so that the number of video tokens and action tokens remains balanced—for example, by setting the video frame rate to one-sixth of the action frame rate.

Experts Details. For the generative expert, we employ Wan 2.2 5B [42] as the video foundation model for its accessibility and ease of use. We extend its self-attention context to create a cross-modal Tri-model Joint Attention mechanism. For the action expert, we construct a Transformer block of the same depth as Wan. Each block comprises AdaLN for injecting rectified flow timesteps, a Feed-Forward Network (FFN), and the Tri-model Joint Attention for cross-expert interaction. We select Qwen3-VL-2B [2, 3, 44] for our understanding expert due to its inherent capabilities in 3D grounding, spatial understanding, and precise object localization, which are crucial for robotic manipulation. The input to this expert is taken from the last-layer corresponding tokens of the VLM. The understanding expert itself consists of several Transformer blocks, each containing Layer Normalization, an FFN, and the Tri-model Joint Attention.

4.2. Latent Actions

We further address Challenge 2 to leverage large-scale heterogeneous data by learning generalizable action patterns directly from visual dynamics. Specifically, we introduce **latent actions** that encode the motion learned directly from pixels. These latent actions allow the model to absorb motion knowledge from various sources such as internet videos, egocentric human demonstrations, and multi-robot trajectories, thereby strengthening the pretraining of action expert even on data without explicit action labels.

Optical Flow Based Representation. We adopt optical flow as a natural representation of motion, which captures pixel-level displacements between consecutive frames. Specifically, optical flows are computed by DPFlow [33] and then converted into RGB images. To compress this high-dimensional representation into a control-level space, we employ a deep convolutional variational autoencoder (DC-AE [13]) that reconstructs the flow while encoding it into four 512-dimensional tokens. A lightweight encoder then projects these concatenated 4×512 features into a 14-dimensional vector, roughly matching the scale of typical robot action spaces. The overall architecture is shown in Figure 3. This dimensional correspondence ensures that the latent representation can align naturally with real robotic controls and act as a bridge between perception and action.

Training and Distribution Alignment. To help align the latent space to realistic action space, we incorporate task-agnostic data following AnyPos [39]. Specifically, task-agnostic data uses Curobo to collect image-action pairs by

randomly sampling the target robot’s action space in a task-agnostic manner. This data provides additional real action supervision, helping the VAE learn an embedding that reflects feasible motor behaviors and anchors the latent actions to the true control distribution.

During training, we mix 90% unlabeled data for self-supervised reconstruction with 10% labeled trajectories for weak action supervision, where the labeled portion includes both task-agnostic data and standard robot demonstrations. Dimensional correspondence and weak action supervision jointly drive the latent-action distribution to align with the real action distribution, allowing motion priors learned from videos to naturally map to executable controls.

The total loss combines reconstruction, alignment, and KL regularization:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_a \|a_{\text{real}} - a_{\text{pred}}\|^2 + \beta \mathcal{L}_{\text{KL}}, \quad (2)$$

where $\mathcal{L}_{\text{recon}}$ minimizes flow-reconstruction error, the second term aligns latent and real actions, $\mathcal{L}_{\text{KL}}$ regularizes the latent space; λ_a and β are hyperparameters.

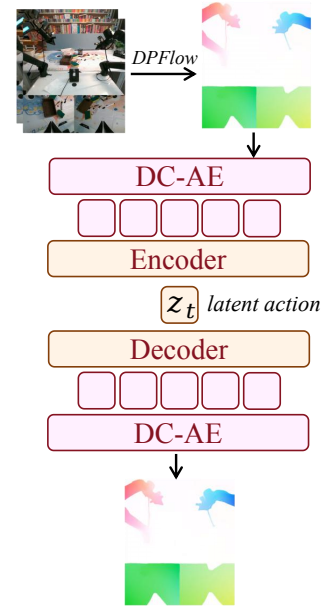


Figure 3. The Latent Action VAE.

4.3. Model Training and Data

Motus Training. Motus is trained in three structured stages (Tab. 1) to progressively integrate physical interaction priors from diverse datasets into a policy transferable to a target robot. Each stage addresses a key challenge:

- **Stage 1: Learning Visual Dynamics.** To anchor the model in realistic physical interactions, we first adapt the Video Generation Model (VGM) using multi-robot trajectories and human videos. This enables the VGM to

generate plausible future video sequences of tasks from a language instruction and an initial image.

- **Stage 2: Learning Action Representations.** To bridge visual forecasts with control, we pretrain the entire Motus model (VLM frozen) on videos, language, and latent actions. This stage initializes the action expert by embedding knowledge of motion and interaction into the latent action space.
- **Stage 3: Specializing for the Target Robot.** We finalize the model by fine-tuning it on target-robot data, ensuring that the acquired priors are fully adapted to the specific embodiment’s dynamics and kinematics.

Table 1. **Motus Training.**

Stage	Data	Training
Pretrained Foundation Models (Off-the-shelf)	Level 1: Web Data	VGM and VLM
Stage 1 (Video Generation)	Level 2: Egocentric Human Videos Level 3: Synthetic Data Level 5: Multi-Robot Task Trajectory Data	Only VGM
Stage 2 (Unified Training with Latent Actions)	Level 2: Egocentric Human Videos Level 3: Synthetic Data Level 4: Task-agnostic Data Level 5: Multi-Robot Task Trajectory Data	Motus (all 3 experts, with latent actions)
Stage 3 (SFT)	Level 6: Target-Robot Task Trajectory Data	Motus (all 3 experts, with actions)

Data. To equip robots with generalizable manipulation skills, we leverage large-scale multimodal data that encapsulates rich prior knowledge—from semantic understanding and physical reasoning to spatiotemporal dynamics and decision-making. As outlined in Section 3, embodied data inherently spans multiple modalities: language ℓ , image $\mathbf{o}$, and action $\mathbf{a}$ ¹. By considering the presence or absence of each modality, we systematically identify all meaningful data types²:

- **Language + Image + Action:** robot trajectories (e.g., used in VLAs), $\{\ell, \mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$.
- **Language + Image:** video sequences $\{\ell, \mathbf{o}_1, \dots, \mathbf{o}_N\}$ or image-text pairs $\{(\mathbf{o}, \ell)\}$.

¹In joint position control, proprioception and action share the same representation space.

²Language can be annotated post-hoc to support task-oriented learning.

- **Image + Action:** task-agnostic interaction data $\{(\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_i, \mathbf{a}_i)\}$.
- **Language-only:** textual corpora $\{\ell\}$.

We exclude data lacking visual modality (e.g., language + action) as it is unsuitable for visuomotor policy learning. The remaining types form the complete spectrum of useful sources for embodied policy acquisition. To structure this diversity, we introduce the *embodied data pyramid* (Fig. 4), which organizes data types hierarchically by richness and policy relevance.

Our framework effectively integrates and aligns all six data levels—from large-scale but indirect web sources to targeted robot demonstrations—across tailored training stages (Tab. 1), unifying heterogeneous datasets [1, 14, 24, 31, 48] within a single, cohesive model architecture.

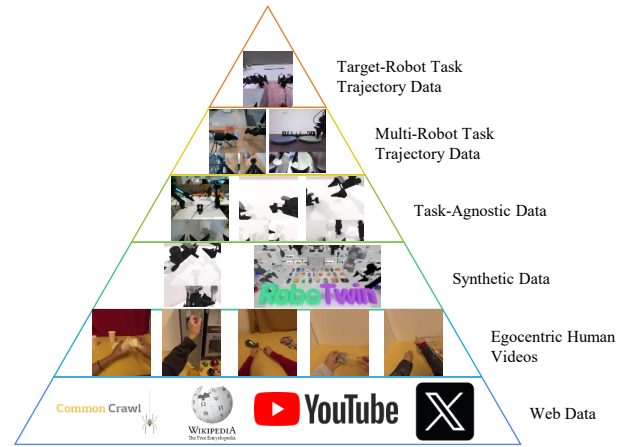


Figure 4. The Embodied Data Pyramid categorizes data into six levels, from Level 1 at the base to Level 6 at the top. Data quantity decreases from bottom to top, while data quality increases. The order of Levels 3 and 4 may sometimes vary.

5. Experiments

We conduct extensive experiments to assess the effectiveness of Motus in both simulated and real-world environments.

5.1. Baselines

We compare Motus against several state-of-the-art methods: $\pi_{0.5}$ [8] and X-VLA [60]. We evaluate all the models in simulation environments and further assess the performance of the baseline model $\pi_{0.5}$ in real-world tasks. We also compared both the from-scratch and Stage-1-only trained models against our own model.

5.2. Evaluation in Simulation Environment

We evaluated single-task performance on 50 representative manipulation tasks from the RoboTwin 2.0 tasks in randomized scenes. To probe the general ability of our method, we carry out multi-task training: Motus and all baselines are

Table 2. Evaluation on RoboTwin 2.0 Simulation (Clean vs Randomized, 50+ tasks).

Simulation Task	$\pi_{0.5}$		X-VLA		w/o Pretrain		Stage1		Motus	
	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
<i>Place Dual Shoes</i>	12%	7%	79%	88%	78%	80%	94%	94%	93%	87%
<i>Move Stapler Pad</i>	16%	18%	78%	73%	49%	37%	75%	68%	83%	85%
<i>Stack Blocks Two</i>	48%	56%	92%	87%	96%	94%	99%	99%	100%	98%
<i>Scan Object</i>	42%	38%	14%	36%	42%	50%	56%	69%	67%	66%
<i>Place Object Stand</i>	74%	65%	86%	88%	91%	93%	93%	96%	98%	97%
<i>Place Fan</i>	25%	36%	80%	75%	77%	85%	77%	85%	91%	87%
<i>Move Pillbottle Pad</i>	33%	29%	73%	71%	83%	83%	96%	90%	93%	96%
<i>Pick Dual Bottles</i>	10%	6%	47%	36%	58%	68%	7%	17%	96%	90%
<i>Blocks Ranking Rgb</i>	43%	35%	83%	83%	92%	88%	97%	98%	99%	97%
.....(50 tasks)										
<i>Turn Switch</i>	5%	6%	40%	61%	69%	60%	59%	64%	84%	78%
<i>Pick Diverse Bottles</i>	5%	3%	58%	36%	53%	62%	18%	18%	90%	91%
<i>Place Bread Basket</i>	48%	56%	81%	71%	73%	83%	89%	87%	91%	94%
<i>Stack Blocks Three</i>	15%	16%	6%	10%	71%	76%	99%	95%	91%	95%
<i>Put Bottles Dustbin</i>	12%	9%	74%	77%	36%	33%	34%	24%	81%	79%
<i>Place Can Basket</i>	19%	25%	49%	52%	46%	62%	66%	55%	81%	76%
<i>Stamp Seal</i>	36%	23%	76%	82%	80%	88%	93%	95%	93%	92%
<i>Hanging Mug</i>	3%	3%	23%	27%	14%	10%	37%	25%	38%	38%
<i>Handover Block</i>	18%	19%	73%	37%	34%	15%	55%	55%	86%	73%
<i>Stack Bowls Three</i>	33%	35%	76%	86%	90%	74%	86%	83%	79%	87%
<i>Place Object Basket</i>	43%	36%	44%	39%	74%	75%	76%	80%	81%	87%
<i>Open Microwave</i>	35%	37%	79%	71%	83%	82%	82%	84%	95%	91%
Average (%)	42.98	43.84	72.80	72.84	72.8	77.00	82.86	81.86	88.66	87.02

trained on 2500 demonstrations collected in clean scenes (50 per task) plus 25000 demonstrations gathered in heavily randomized scenes (500 per task). The randomization includes random backgrounds, a cluttered table, table-height perturbations, and randomized lighting. All models are finetuned for 40k steps on the RoboTwin dataset starting from their pretrained checkpoints, and we evaluate performance by measuring the success rate of each task over 100 execution trials.

This benchmark is particularly challenging and informative because it contains a large variety of task scenes and randomized instructions, testing a model’s ability to handle various manipulation settings. Its strong background and environmental variability further evaluate the generalization under distribution shift. Moreover, all models are allowed only 40k finetuning steps on top of their pretrained checkpoints, providing a strict and fair assessment of the effectiveness of different pretraining strategies.

As shown in Tab. 2, Motus achieves state-of-the-art performance on the RoboTwin 2.0 randomized multi-task setting, delivering over a 45% absolute improvement compared with the $\pi_{0.5}$ model. By using a unified MoT model, Motus successfully integrates vision, language, and action generation, solving **Challenge 1**. In **Challenge 2**, the introduction of latent actions enables Motus to effectively leverage both labeled and large-scale unlabeled data, improving generalization across embodiments and capturing rich motion priors. This combination of techniques allows Motus to overcome the limitations of previous approaches and achieve superior

performance.

5.3. Real-World Experiments

We evaluate Motus across two distinct real-world dual-arm robotic platforms, AC-One and Agilex-Aloha-2 under a comprehensive set of non-trivial tasks that span various dimensions of policy capabilities including: (1) Spatial Understanding (2) Deformable Objects Manipulation (3) Precision Fluid Control (4) Visual understanding (5) Long-Horizon Planning, such as fold towel, brew coffee using drip coffee machine and grind coffee beans with grinder.

For each task, we employed 100 trajectories for training. Consistent with the simulator, a multi-task joint training scheme was adopted: all tasks on each robotic platform were trained collectively within a single model, which was subsequently evaluated on every individual task. This approach provides a comprehensive and rigorous assessment of the model’s robustness and generalization capabilities.

We choose $\pi_{0.5}$ as our baseline. Since most tasks involve long-horizon reasoning and are decomposable, we employed the partial success rate for evaluation. This metric quantifies performance by decomposing a task into subtasks, where the model earns partial scores for achieving specific subgoals and a full score only for overall success, thereby offering a more compelling demonstration of its capability. Examples are shown in Table 6 and Table 5.

The results are reported in Table 3. Our results demonstrate that Motus significantly outperforms the baseline $\pi_{0.5}$ across all tasks on both robotic arms. Visualizations are

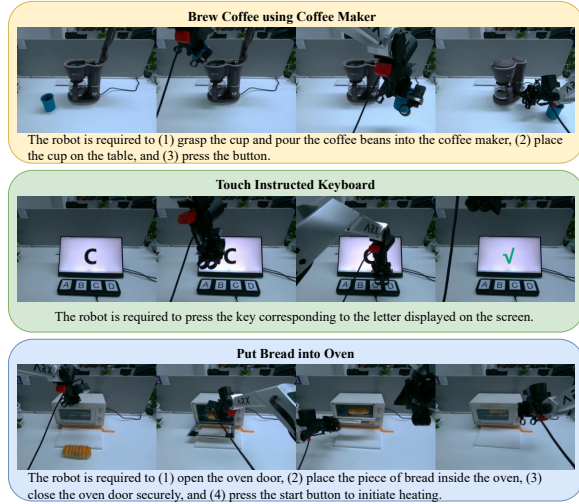


Figure 5. **Task Definitions and Visualizations.** For each task, we describe its language instruction and definitions of each sub-task.

Table 3. Robotic Manipulation Tasks Performance Across Platforms (Partial Success Rate %).

Task Description	$\pi_{0.5}$	w/o Pretrain	Motus
AC-One			
Fold Towel	4	1	14.5
Brew Coffee using Coffee Maker	0	0	62
Get Water from Water Dispenser	30	8	36
Place Cube into Plate	46	60	100
Place Cube into Plate(OOD)	28.125	18.75	75
Grind Coffee Beans with Grinder	8	0	92
Pour Water from Kettle to Flowers	5	5	65
Touch Instructed Keyboard	0	100	82.5
Put Bread into Oven	12	40	42
Average	14.79	25.86	63.22
Agilex-Aloha-2			
Fold Towel	27.5	0	39
Get Water from Water Dispenser	62	8	96
Pour Water from Kettle to Flowers	45	40	47.5
Touch Instructed Keyboard	72.5	85	80
Put Bread into Oven	36	0	34
Average	48.60	26.60	59.30

provided in Figure 5

5.4. Ablation Study

We performed ablation studies to demonstrate the contribution of each training stage. This involved benchmarking models without pretraining and only Stage 1 pretraining. Evaluations were carried out in the RoboTwin 2.0 simulator to measure accuracy. In real-world deployments we compare Motus against its from-scratch counterpart. The results in simulator are summarized in Fig 6, and results in real-world

Table 4. Put Bread into Oven Task on AC-One Platform with a Detailed Subtask Breakdown. The number preceding each subtask indicates the score assigned to its successful completion.

Subgoal	$\pi_{0.5}$	w/o Pretrain	Motus
0.0: Complete Failure	6	4	5
0.2: Open the Oven	3	0	0
0.4: Grab the Bread	0	2	1
0.6: Put the Bread into the Oven	1	1	0
0.8: Close the Oven	0	2	1
1.0: Spin the Button	0	1	3
Partial Success Rate	12%	40%	42%

Table 5. Get Water from Water Dispenser Task on Agilex-Aloha-2 Platform with a Detailed Subtask Breakdown. The number preceding each subtask indicates the score assigned to its successful completion.

Subgoal	$\pi_{0.5}$	w/o Pretrain	Motus
0.0: Complete Failure	0	8	0
0.4: Grab the cup	5	2	0
0.8: Fill the cup with water	4	0	2
1.0: Complete Success	1	0	8
Partial Success Rate	62%	8%	96%

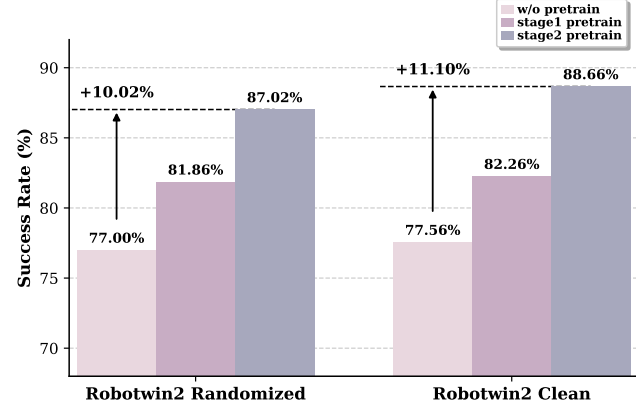


Figure 6. **Ablation in RoboTwin 2.0 Randomized Multi-task Setting.** The figure presents the total success rates (%) of the original Motus (Stage 2 Pretrain) and its two variants: Without Pretrain and Stage 1 Pretrain.

experiments are shown in Table 3.

6. Conclusion and Limitations

In this work, we present Motus, a unified latent-action world model that integrates mainstream capabilities of embodied foundation models into a single generative framework, *i.e.*, vision-language understanding, video generation, inverse dynamics, world modeling, and video-action joint prediction. By connecting pretrained experts through MoT, coordinating multimodal modeling with a UniDiffuser-style scheduler, and introducing latent actions as a pixel-level “delta action” and motion representation, Motus effectively learns from large-scale heterogeneous data and inherits both gen-

eral multimodal priors and rich physical interaction knowledge. Extensive experiments across simulation and real-world environments demonstrate that Motus consistently outperforms existing state-of-the-art embodied models (improved by **+15~45%** in simulation and **+11~48%** in real-world scenarios), validating the importance of unifying multimodal generative capabilities and shared motion priors. We hope Motus inspires future research on unified architectures, motion-centric representation learning, and large-scale embodied pretraining.

In the future, we will continue to explore more advanced unified model architectures, pursue more universal motion priors, and learn latent actions from internet-scale general videos for embodied intelligence.

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Motus: A Unified Latent Action World Model

Supplementary Material

7. Training and Inference of the Unified Model

In this section, we analyze the training and inference procedures of the unified model, from both theoretical and experimental perspectives.

7.1. Theoretical Analysis

During each training iteration, given $o_{t:t+k}^0$ and $a_{t:t+k}^0$, Motus samples different timesteps τ_o, τ_a and noise ϵ_o, ϵ_a for them respectively, construct the interpolated trajectories $o_{t+1:t+k}^{\tau_o}, a_{t+1:t+k}^{\tau_a}$ based on rectified flow, and compute the loss between the predicted velocity field v_o^θ, v_a^θ and its ground truth v_o, v_a obtained by path differentiation with t .

Algorithm 1 Training

```

1: repeat
2:    $o_{t:t+k}^0, a_{t+1:t+k}^0, \ell \sim D_{expert}$ 
3:    $\tau_o, \tau_a \sim \text{Uniform}(\{1, 2, \dots, T_\tau\})$ 
4:    $\epsilon_o, \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:    $o_{t+1:t+k}^{\tau_o} = (1 - \tau_o)o_{t+1:t+k}^0 + \tau_o\epsilon_o$ 
6:    $a_{t+1:t+k}^{\tau_a} = (1 - \tau_a)a_{t+1:t+k}^0 + \tau_a\epsilon_a$ 
7:    $v_o^\theta, v_a^\theta = \text{Model}_\theta(o_t^0, o_{t+1:t+k}^{\tau_o}, a_{t+1:t+k}^{\tau_a}, \tau_o, \tau_a, \ell)$ 
8:    $l_{\text{action}}^\theta = \|v_a^\theta - (\epsilon_a - a_{t+1:t+k}^0)\|_2^2$ 
9:    $l_{\text{obs}}^\theta = \|v_o^\theta - (\epsilon_o - o_{t+1:t+k}^0)\|_2^2$ 
10:   $l^\theta = l_{\text{action}}^\theta + l_{\text{obs}}^\theta$ 
11:   $\theta \leftarrow \theta - \eta \nabla_\theta l^\theta$ 
12: until converged

```

During inference, Motus can switch between the following five different modes.

VGM. To enable VGM $p(o_{t+1:t+k}^0 | o_t^0, \ell)$, given o_t^0 and ℓ as conditions, we set the starting timesteps for both the observations and actions to T_τ , randomly sample $\epsilon_a, \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then apply Alg. 2 to gradually infer $o_{t+1:t+k}^0$ from ϵ_o , while keeping $a_{t+1:t+k}^{T_\tau}$ consistently noisy as ϵ_a .

Algorithm 2 VGM

```

Require:  $o_t^0, \ell, \theta$ 
1:  $\epsilon_o, \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2:  $o_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_o$ 
3:  $a_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_a$ 
4: for  $\tau = T_\tau \dots 1$  do
5:    $v_o, v_a = \text{Model}_\theta(o_t^0, o_{t+1:t+k}^\tau, a_{t+1:t+k}^{T_\tau}, \tau, T_\tau, \ell)$ 
6:    $o_{t+1:t+k}^{\tau-1} = o_{t+1:t+k}^\tau + v_o d\tau$ 
7: end for
8: return  $o_{t+1:t+k}^0$ 

```

World Model. To enable world model $p(o_{t+1:t+k}^0 | o_t^0, a_{t+1:t+k}^0)$, given o_t^0 and $a_{t+1:t+k}^0$ as conditions, we set the starting timesteps for the observations and actions to T_τ and 0 respectively, randomly sample $\epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then apply Alg. 3 to gradually infer $o_{t+1:t+k}^0$ from ϵ_o , while keeping $a_{t+1:t+k}^0$ always clean.

Algorithm 3 World Model

```

Require:  $o_t^0, a_{t+1:t+k}^0, \ell, \theta$ 
1:  $\epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2:  $o_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_o$ 
3: for  $\tau = T_\tau \dots 1$  do
4:    $v_o, v_a = \text{Model}_\theta(o_t^0, o_{t+1:t+k}^\tau, a_{t+1:t+k}^0, \tau, 0, \ell)$ 
5:    $o_{t+1:t+k}^{\tau-1} = o_{t+1:t+k}^\tau + v_o d\tau$ 
6: end for
7: return  $o_{t+1:t+k}^0$ 

```

IDM. To enable IDM $p(a_{t+1:t+k}^0 | o_{t:t+k}^0)$, given $o_{t:t+k}^0$ as conditions, we set the starting timesteps for the observations and actions to 0 and T_τ respectively, randomly sample $\epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then apply Alg. 4 to gradually infer $a_{t+1:t+k}^0$ from ϵ_a , while keeping $o_{t:t+k}^0$ always clean.

Algorithm 4 IDM

```

Require:  $o_{t:t+k}^0, \ell, \theta$ 
1:  $\epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2:  $a_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_a$ 
3: for  $\tau = T_\tau \dots 1$  do
4:    $v_o, v_a = \text{Model}_\theta(o_{t:t+k}^0, a_{t+1:t+k}^\tau, 0, \tau, \ell)$ 
5:    $a_{t+1:t+k}^{\tau-1} = a_{t+1:t+k}^\tau + v_a d\tau$ 
6: end for
7: return  $a_{t+1:t+k}^0$ 

```

VLA. To enable VLA $p(a_{t+1:t+k}^0 \mid o_t^0, \ell)$, given o_t^0 and ℓ as conditions, we set the starting timesteps for both the observations and actions to T_τ , randomly sample $\epsilon_a, \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then apply Alg. 5 to gradually infer $a_{t+1:t+k}^0$ from ϵ_a , while keeping $o_{t+1:t+k}^{T_\tau}$ consistently noisy as ϵ_o .

Algorithm 5 VLA

Require: o_t^0, ℓ, θ

- 1: $\epsilon_o, \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: $o_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_o$
- 3: $a_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_a$
- 4: **for** $\tau = T_\tau \dots 1$ **do**
- 5: $v_o, v_a = \text{Model}_\theta(o_t^0, o_{t+1:t+k}^{T_\tau}, a_{t+1:t+k}^{T_\tau}, T_\tau, \tau, \ell)$
- 6: $a_{t+1:t+k}^{\tau-1} = a_{t+1:t+k}^{T_\tau} + v_a d\tau$
- 7: **end for**
- 8: **return** $a_{t+1:t+k}^0$

Video-Action Joint Prediction Model. To enable video-action joint prediction model $p(o_{t+1:t+k}^0, a_{t+1:t+k}^0 \mid o_t^0, \ell)$, given o_t^0 and ℓ as conditions, we set the starting timesteps for both the observations and actions to T_τ , randomly sample $\epsilon_a, \epsilon_o \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then apply Alg. 2 to gradually infer $a_{t+1:t+k}^0$ from ϵ_a and $o_{t+1:t+k}^0$ from ϵ_o .

Algorithm 6 Video-Action Joint Prediction Model

Require: o_t^0, ℓ, θ

- 1: $\epsilon_o, \epsilon_a \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: $o_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_o$
- 3: $a_{t+1:t+k}^{T_\tau} \leftarrow \epsilon_a$
- 4: **for** $\tau = T_\tau \dots 1$ **do**
- 5: $v_o, v_a = \text{Model}_\theta(o_t^0, o_{t+1:t+k}^{T_\tau}, a_{t+1:t+k}^{T_\tau}, \tau, \tau, \ell)$
- 6: $o_{t+1:t+k}^{\tau-1} = o_{t+1:t+k}^{T_\tau} + v_o d\tau$
- 7: $a_{t+1:t+k}^{\tau-1} = a_{t+1:t+k}^{T_\tau} + v_a d\tau$
- 8: **end for**
- 9: **return** $o_{t+1:t+k}^0, a_{t+1:t+k}^0$

7.2. Experimental Results

VGM. As shown in Fig. 7 and Fig. 9, when Motus performs in VGM mode, it shows high-quality visualization results across both Agilex-Aloha-2 and AC-One embodiments, demonstrating the strong video generation capabilities.

World Model. As shown in Fig. 11, Fig. 10 and Tab. 6, when Motus performs in world model mode, it shows high-quality video generation results across two embodiments on real-world robot data, demonstrating strong future prediction capabilities.

Table 6. **Generative Quality of Motus in World Model Mode.** The metrics were evaluated on real-world robot data across two robotic platform.

Platform	FID↓	FVD↓	SSIM↑	LPIPS↓	PSNR↑
Agilex-Aloha-2	9.4571	49.2848	0.88618	0.05449	26.1021
AC-One	12.9609	73.1325	0.84605	0.07280	24.0379
Avg.	11.209	61.20865	0.8661	0.063645	25.0700

IDM. To validate the effectiveness of our model as an IDM, we trained two baseline IDMs for comparison: one based on a pretrained ResNet-18 backbone followed by an MLP layer, and another using DINOv2 features with an MLP head. Both models were trained on the RobotWin 2.0 randomized dataset using the Agilex-Aloha-2 robotic platform. Each model takes the current observation as input and predicts a sequence of future actions with an action chunk size of 16, which is consistent with the configuration used by Motus in RobotTwin. The training objective was to minimize the Mean Squared Error (MSE) between predicted and ground-truth actions.

As shown in Table 7, when Motus performs in IDM mode, it achieves a lower action MSE than the specifically trained IDM baselines. This indicates that our model not only serves as an effective policy but also excels at inverse dynamics modeling, even outperforming models explicitly trained for that purpose.

Table 7. **Action MSE of IDM.** The models are tested on 100 samples of RoboTwin 2.0 randomized data.

ResNet18+MLP	DINOv2+MLP	Motus
0.044	0.122	0.014

VLA. As shown in Tab. 8, when Motus performs in the VLA mode, it also demonstrates competitive performance on RoboTwin 2.0 randomized data compared to the video-action joint prediction mode.

Table 8. **Average Success Rate on RoboTwin 2.0 Randomized Data of VLA.**

Motus (VLA)	Motus (Joint)
83.90	87.02

Video-Action Joint Prediction Model. As shown in Fig. 12, when Motus performs in the video-action joint prediction model mode, it demonstrates strong capabilities in generating both videos and precise actions simultaneously.



Figure 7. Visualization of Motus’s VGM mode on Agilex-Aloha-2.

8. More Experiments Results

8.1. Overall Comparison on RoboTwin 2.0 Simulation Data with More Baselines

Tab. 14 shows the evaluation results on RoboTwin 2.0 Simulation, presenting the performance of Motus and other baselines on all 50 tasks under both clean scenes and randomized scenes.

8.2. Other Benchmarks

LIBERO-Long. LIBERO-Long is the long-horizon subset of the LIBERO benchmark, comprising 10 language-conditioned manipulation tasks from LIBERO-100 that require multi-stage decision making, diverse manipulation skills, and robust knowledge transfer across objects and scenes. Under the standard LIBERO-Long evaluation protocol, our method achieves an average success score of **97.6**, matching the best reported performance of X-VLA and thereby reaching state-of-the-art results on this benchmark.

π_0	GR00T-N1	UniVLA	OpenVLA-OFT	X-VLA	Motus
85.2	90.6	94.0	94.5	97.6	97.6

Table 9. Evaluation on LIBERO-Long Benchmark

VLABench. VLABench is an open-source benchmark for evaluating universal language-conditioned manipulation task learning, covering multiple dimensions such as manipulation skills, vision understanding, semantic comprehension, common sense, and reasoning. A single Motus model was fine-tuned on multiple tasks and subsequently evaluated based on its success rate across 3 tasks on 2 tracks provided by VLABench: In Distribution and Cross Category. The result is shown in Tab. 10. The evaluation result of $\pi_{0.5}$ is sourced from its official implementation.

8.3. More Real-World Results

Fig. 8 illustrates the visualization of the Motus execution for each task presented in Tab. 3. The detailed results containing subtask breakdown of the real-world tasks on the AC-One and Agilex-Aloha-2 platforms are presented in Tab. 15 and

Model	Add Condiment	Select Toy	Select Fruit	Avg.
In Distribution				
$\pi_{0.5}$	0.56	0.3	0.42	0.43
Motus	0.63	0.47	0.33	0.48
Cross Category				
$\pi_{0.5}$	0.06	0.24	0.36	0.22
Motus	0.14	0.40	0.20	0.25

Table 10. Evaluation of Success Rate on VLABench

Component	Configuration
Action Expert	
Hidden Size	1024
Layers	30
Attention Heads	24
Layer Norm Epsilon	1e-5
Activation Function	GELU
Understand Expert	
Hidden Size	512
Layers	30
Attention Heads	24
Layer Norm Epsilon	1e-5
Activation Function	GELU
Latent Action VAE	
λ_a (Action Alignment)	1.0
β (KL Regularization)	1×10^{-6}
Sampling Rate	
Video Frames	8 @ 5Hz
Action Chunk	48 @ 30Hz
Flow Matching	
Inference Steps	10
Sampling Strategy	Logit Normal
Model Scale	
VGM	5.00B
VLM	2.13B
Act. Expert	641.5M
Und. Expert	253.5M
Total	8B

Table 11. Motus architecture hyperparameters and key configuration settings.

Tab. 16. The number preceding each subtask indicates the score assigned to its successful completion. For the towel-folding task, we evaluate each towel type four times. For the grab-cube task, we evaluate each cube type five times for both the in-domain and out-of-domain settings.

9. Implementation Details

9.1. Model Architecture

Tab. 11 provides the key hyperparameter settings for the Motus model architecture.

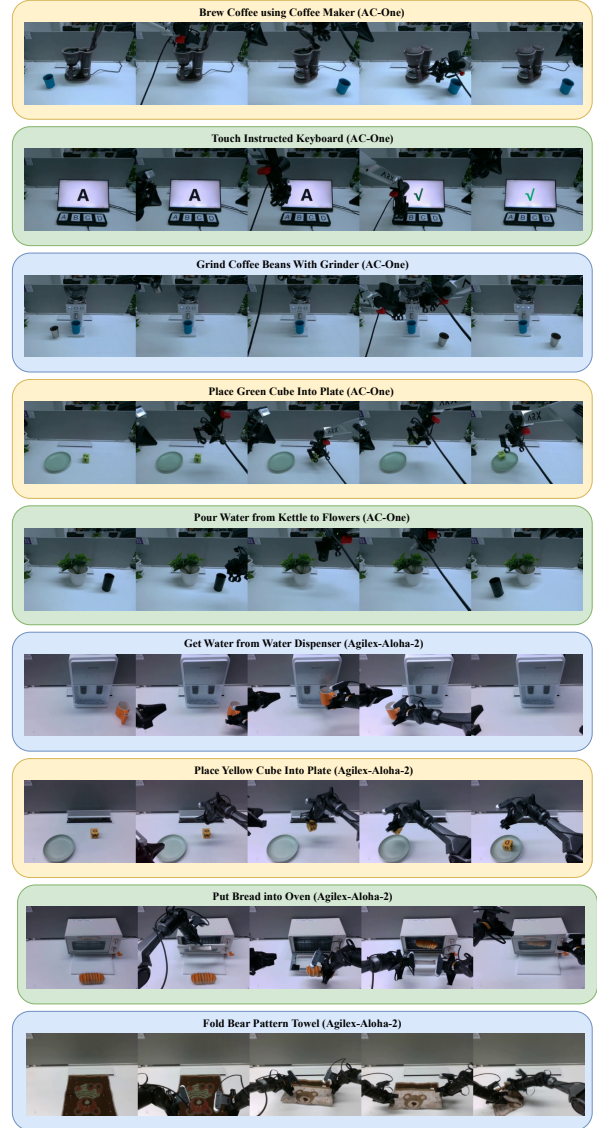


Figure 8. Demonstrations of Motus for real-world tasks execution featuring 2 robots and 9 tasks.

9.2. Datasets

Tab. 12 shows the training data of Motus.

9.3. Training Configuration

Tab. 13 provides the detailed training configuration for the three stages of Motus.

Table 12. Detailed information about pre-training and fine-tuning datasets.

Dataset	Size	Embodiment	Data Level in the Pyramid
Egodex [24]	230,949	Human	Level 2: Egocentric Human Videos
Agibot [1]	728,209	Genie-1 Robot	Level 5: Multi-Robot Task Trajectory Data
RDT [31]	6,083	Aloha Robot	Level 5: Multi-Robot Task Trajectory Data
RoboMind Franka [48]	9,589	Franka Robot	Level 5: Multi-Robot Task Trajectory Data
RoboMind Aloha [48]	7,272	Aloha Robot	Level 5: Multi-Robot Task Trajectory Data
RoboTwin [14]	27,500	Aloha Robot	Level 3: Synthetic Data
Task-Agnostic Data [39]	1,000	Aloha Robot	Level 4: Task-Agnostic Data
In-house Data	2,000	Aloha Robot	Level 6: Target-Robot Task Trajectory Data

Table 13. Training Configuration across Three Stages.

Stages	Stage 1	Stage 2	Stage 3
Batch Size	256	256	256
Learning Rate	8×10^{-5}	5×10^{-5}	$1 \sim 5 \times 10^{-5}$
Optimizer	AdamW	AdamW	AdamW
Weight Decay	0.01	0.01	0.01
GPU Hours	~8000	~10000	~400

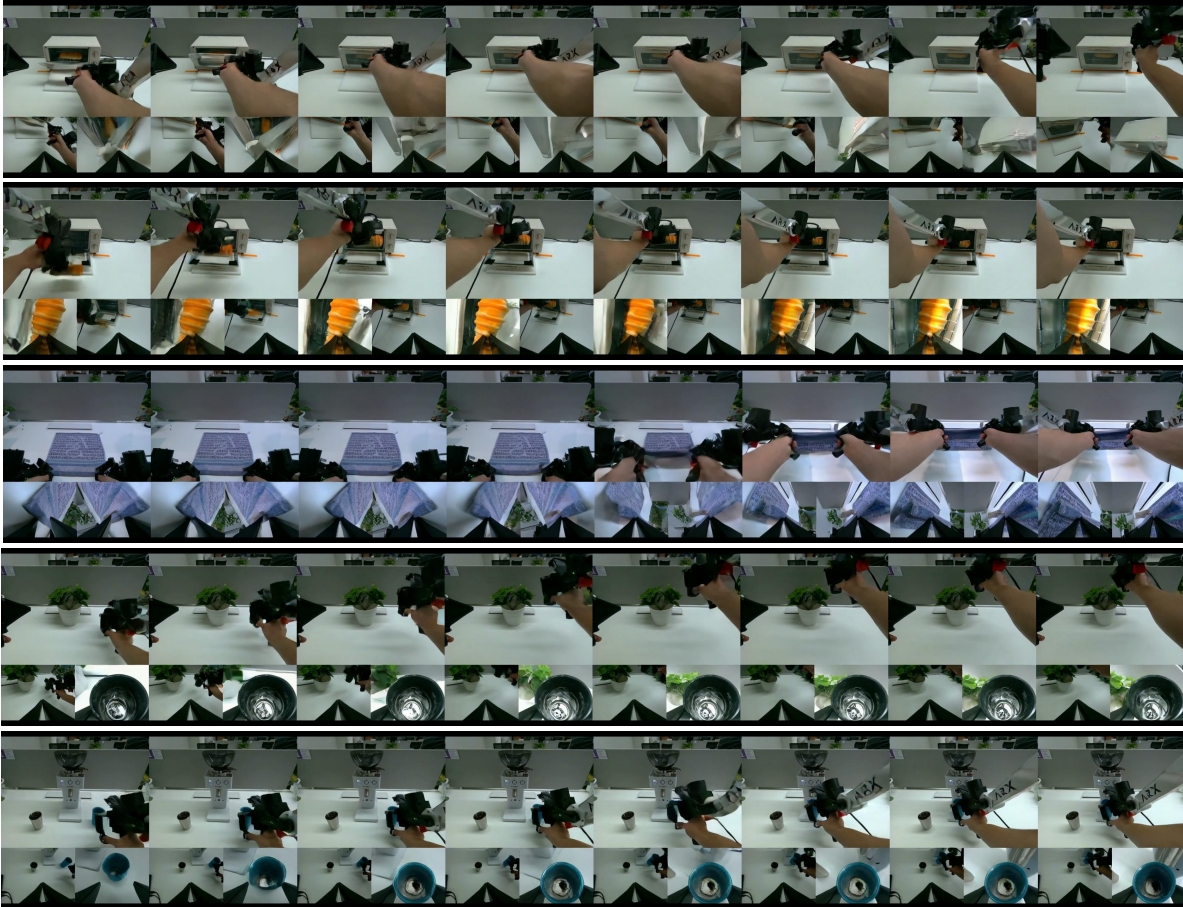


Figure 9. Visualization of Motus’s VGM mode on AC-One.

Table 14. Evaluation on RoboTwin 2.0 Simulation (Clean vs Randomized, 50+ tasks).

Simulation Task	GO-1		$\pi_{0.5}$		X-VLA		w/o Pretrain		Stage1		Motus	
	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.	Clean	Rand.
Adjust Bottle	49%	62%	79%	83%	100%	99%	99%	97%	98%	94%	89%	93%
Beat Block Hammer	6%	10%	63%	50%	92%	88%	88%	90%	88%	82%	95%	88%
Blocks Ranking Rgb	7%	3%	43%	35%	83%	83%	92%	88%	97%	98%	99%	97%
Blocks Ranking Size	2%	2%	8%	14%	67%	74%	38%	50%	73%	68%	75%	63%
Click Alarmclock	95%	90%	97%	93%	99%	99%	100%	99%	100%	100%	100%	100%
Click Bell	98%	95%	75%	76%	100%	100%	100%	100%	100%	100%	100%	100%
Dump Bin Bigbin	57%	45%	30%	42%	79%	77%	94%	96%	98%	96%	95%	91%
Grab Roller	99%	99%	90%	89%	100%	100%	100%	100%	100%	100%	100%	100%
Handover Block	9%	12%	18%	19%	73%	37%	34%	15%	55%	55%	86%	73%
Handover Mic	12%	8%	28%	18%	0%	0%	98%	95%	80%	88%	78%	63%
Hanging Mug	0%	0%	3%	3%	23%	27%	14%	10%	37%	25%	38%	38%
Lift Pot	92%	92%	0%	0%	99%	100%	90%	87%	87%	84%	96%	99%
Move Can Pot	16%	4%	29%	27%	89%	86%	43%	53%	56%	65%	34%	74%
Move Pillbottle Pad	9%	11%	33%	29%	73%	71%	83%	83%	96%	90%	93%	96%
Move Playingcard Away	37%	24%	59%	67%	93%	98%	50%	47%	77%	84%	100%	96%
Move Stapler Pad	3%	4%	16%	18%	78%	73%	49%	37%	75%	68%	83%	85%
Open Laptop	65%	60%	19%	35%	93%	100%	89%	89%	91%	96%	95%	91%
Open Microwave	12%	14%	35%	37%	79%	71%	83%	82%	82%	84%	95%	91%
Pick Diverse Bottles	61%	56%	5%	3%	58%	36%	53%	62%	18%	18%	90%	91%
Pick Dual Bottles	81%	74%	10%	6%	47%	36%	58%	68%	7%	17%	96%	90%
Place A2b Left	33%	36%	62%	60%	48%	49%	78%	79%	93%	82%	88%	79%
Place A2b Right	31%	22%	62%	57%	36%	36%	86%	83%	94%	90%	91%	87%
Place Bread Basket	47%	52%	48%	56%	81%	71%	73%	83%	89%	87%	91%	94%
Place Bread Skillet	2%	1%	38%	46%	77%	67%	71%	71%	86%	87%	86%	83%
Place Burger Fries	88%	92%	66%	70%	94%	94%	95%	90%	97%	99%	98%	98%
Place Can Basket	29%	37%	19%	25%	49%	52%	46%	62%	66%	55%	81%	76%
Place Cans Plasticbox	68%	77%	40%	47%	97%	98%	96%	99%	97%	100%	98%	94%
Place Container Plate	73%	70%	71%	78%	97%	95%	97%	100%	98%	98%	98%	99%
Place Dual Shoes	6%	10%	12%	7%	79%	88%	78%	80%	94%	94%	93%	87%
Place Empty Cup	44%	39%	75%	86%	100%	98%	97%	97%	96%	97%	99%	98%
Place Fan	1%	0%	25%	36%	80%	75%	77%	85%	77%	85%	91%	87%
Place Mouse Pad	15%	10%	21%	26%	70%	70%	62%	68%	72%	69%	66%	68%
Place Object Basket	48%	49%	43%	36%	44%	39%	74%	75%	76%	80%	81%	87%
Place Object Scale	26%	27%	40%	49%	52%	74%	84%	83%	88%	93%	88%	85%
Place Object Stand	56%	63%	74%	65%	86%	88%	91%	93%	93%	96%	98%	97%
Place Phone Stand	30%	37%	49%	53%	88%	87%	80%	78%	76%	86%	87%	86%
Place Shoe	15%	13%	57%	61%	96%	95%	95%	92%	100%	99%	99%	97%
Press Stapler	66%	51%	80%	70%	92%	98%	97%	94%	96%	98%	93%	98%
Put Bottles Dustbin	7%	4%	12%	9%	74%	77%	36%	33%	34%	24%	81%	79%
Put Object Cabinet	60%	43%	24%	15%	46%	48%	84%	64%	97%	87%	88%	71%
Rotate Qrcode	22%	9%	47%	56%	34%	33%	80%	60%	91%	79%	89%	73%
Scan Object	1%	2%	42%	38%	14%	36%	42%	50%	56%	69%	67%	66%
Shake Bottle Horizontally	97%	92%	96%	100%	100%	100%	100%	97%	100%	96%	100%	98%
Shake Bottle	97%	93%	91%	100%	99%	100%	100%	96%	99%	97%	100%	97%
Stack Blocks Three	1%	1%	15%	16%	6%	10%	71%	76%	99%	95%	91%	95%
Stack Blocks Two	12%	22%	48%	56%	92%	87%	96%	94%	99%	99%	100%	98%
Stack Bowls Three	4%	7%	33%	35%	76%	86%	90%	74%	86%	83%	79%	87%
Stack Bowls Two	51%	45%	78%	66%	96%	93%	98%	98%	97%	98%	98%	98%
Stamp Seal	19%	13%	36%	23%	76%	82%	80%	88%	93%	95%	93%	92%
Turn Switch	34%	30%	5%	6%	40%	61%	69%	60%	59%	64%	84%	78%
Average (%)	37.8	36.24	42.98	43.84	72.8	72.84	77.56	77.00	82.26	81.86	88.66	87.02

Table 15. Real-World Tasks on AC-One Platform with a Detailed Subtask Breakdown.

Subgoal	$\pi_{0.5}$	w/o Pretrain	Motus
Fold Towel			
Types: bear-pattern/blue-yellow/purple/red-blue/pink			
0.0: Complete Failure	16	19	13
0.2: Grab both sides	4	1	3
0.5: One fold complete	-	-	3
0.8: Grab the right side	-	-	1
1.0: Two folds complete	-	-	-
Partial Success Rate	4%	1%	14.5%
Grab Cube			
Types: red/orange/green/yellow			
0.0: Complete Failure	7	8	-
0.5: Grab cube	3	-	-
1.0: Put cube into plate	10	12	20
Partial Success Rate	57.5%	60%	100%
Grab Cube			
OOD setting: cube placed outside training space			
0.0: Complete Failure	11	13	4
0.5: Grab cube	1	-	-
1.0: Put cube into plate	4	3	12
Partial Success Rate	28.125%	18.75%	75%
Brew Coffee using Drip Coffee Machine			
0.0: Complete Failure	10	10	2
0.2: Grab the blue cup	-	-	1
0.5: Pour coffee grounds	-	-	-
0.8: Close the lid	-	-	5
1.0: Turn on the switch	-	-	2
Partial Success Rate	0%	0%	62%
Get Water from Water Dispenser			
0.0: Complete Failure	4	9	4
0.4: Grab the orange cup	5	-	4
0.8: Fill the cup with water	-	1	-
1.0: Put down the cup	1	-	2
Partial Success Rate	30%	8%	36%
Grind Coffee Beans with Grinder			
0.0: Complete Failure	9	10	-
0.3: Grab the metal cup	-	-	-
0.8: Pour the coffee beans	1	-	4
1.0: Press the button	-	-	6
Partial Success Rate	8%	0%	92%
Pour Water from Kettle to Flowers			
0.0: Complete Failure	18	18	4
0.5: Grab the black cup	2	2	6
1.0: Pour water	-	-	10
Partial Success Rate	5%	5%	65%
Touch Keyboard with Hand for Multiple Choice Questions			
0.0: Complete Failure	20	-	3
0.5: Use the correct arm	-	-	1
1.0: Press the right key	-	20	16
Partial Success Rate	0%	100%	82.5%

Table 16. Real-World Tasks on Agilex-Aloha-2 Platform with a Detailed Subtask Breakdown.

Subgoal	$\pi_{0.5}$	w/o Pretrain	Motus
Fold Towel			
Types: bear-pattern/blue-yellow/purple/red-blue/pink			
0.0: Complete Failure	4	20	5
0.2: Grab both sides	11	-	1
0.5: One fold complete	3	-	12
0.8: Grab the right side	1	-	2
1.0: Two folds complete	1	-	-
Partial Success Rate	27.5%	0%	39%
Grab Cube			
Types: red/orange/green/yellow			
0.0: Complete Failure	2	8	-
0.5: Grab cube	1	8	-
1.0: Put cube into plate	17	4	20
Partial Success Rate	87.5%	40%	100%
Grab Cube			
OOD setting: cube placed outside training space			
0.0: Complete Failure	5	13	11
0.5: Grab cube	-	-	-
1.0: Put cube into plate	11	3	5
Partial Success Rate	68.75%	18.75%	31.25%
Put Bread into Oven			
0.0: Complete Failure	5	10	5
0.2: Open the oven	-	-	-
0.4: Grab the bread	1	-	-
0.6: Put the bread into the oven	-	-	3
0.8: Close the oven	4	-	2
1.0: Spin the button	-	-	-
Partial Success Rate	36%	0%	34%
Pour Water from Kettle to Flowers			
0.0: Complete Failure	2	4	3
0.5: Grab the black cup	18	16	15
1.0: Pour water	-	-	2
Partial Success Rate	45%	40%	47.5%
Touch Keyboard with Hand for Multiple Choice Questions			
0.0: Complete Failure	5	-	-
0.5: Use the correct arm	1	6	8
1.0: Press the right key	14	14	12
Partial Success Rate	72.5%	85%	80%

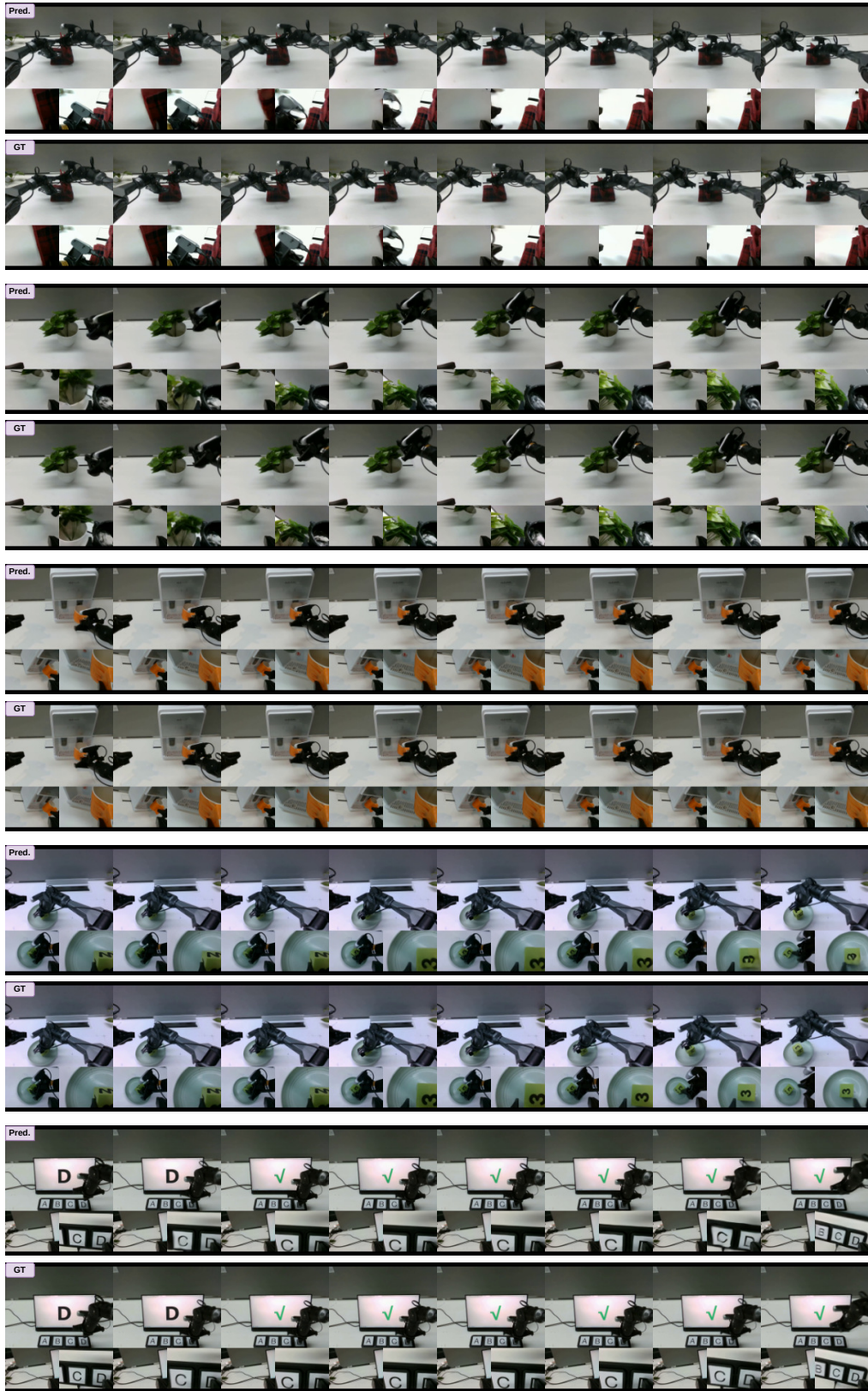


Figure 10. Visualization of Motus's World Model Mode on Agilex-Aloha-2 Dataset.



Figure 11. Visualization of Motus's World Model Mode on AC-One Dataset.



Figure 12. Visualization of Motus’s Video-Action Joint Prediction Model mode during Real-World Inference.